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Department of Computer Science
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Class: B.Sc. (Data Science) Sem-IV
Subject: DS-251-MJ: Relational Database Management System
Question Bank

1. Define concurrent schedule
2. Show compatibility matrix for shared and exclusive lock
3. Give the syntax of GRANT command
4. Define Trigger.
5. What do you mean by time stamp?
6. List various states of a transaction
7. List the disadvantages of concurrent schedules.
8. What is shared memory?
9. What is view?
10. Key-value and graph databases are examples of NoSQL. State TRUE/FALSE.
11. Define cascading rollback.
12. What is shared lock?
13. What do you mean by bound cursor?
14. When does the dirty read occur?
15. Serializability can easily be ensured if access to database is done mutually exclusive manner. State TRUE/FALSE.
16. List any two advantages of shadow paging.
17. Give the syntax of creating view.
18. List any two types of failures.
19. What is a complete schedule ?
20. State any two advantages of RDBMS.
21. What are redo and undo logs?
22. Give any four datatypes used in PL/pgsql.
23. State the atomicity properties of a transaction.
24. What is Parallel Database?

Descriptive Question

1. Write structure of PL/PgSQL code block
2. What do you mean by cascadeless, strict and non recoverable schedule?
3. Discuss the variations of two phase locking protocol
4. What are the reasons of transaction failure?
5. Explain the timestamp ordering protocol for concurrency
6. Write note on shadow paging
7. Explain the distributed database management system.
8. What is trigger ? Explain the syntax to create trigger along with its advantages.
9. Define transaction. Discuss the properties of transaction
10. Explain the concept of locks with multiple granularity
11. What is view? write syntax to create view?
12. What is transaction? Explain ACID property of transaction.
13. Explain timestamp based protocol.
14. What is stored procedure? Give syntax to create stored procedure.
15. Explain cascadeless schedule
16. Explain Log based recovery
17. What is function? Explain with example
18. With suitable diagram explain different states of transaction
19. What is checkpoint? How are they useful in crash recovery
20. Which are different types of log entries are there in system log, explain with examples?
21. Write a short note on multimedia database.
22. What is stored function ? Explain with an example.
23. Explain life cycle of transaction with a diagram.
24. Define view. Explain how to create view with example.
25. Define : (1) (2) (3) Lock Shared Lock Exclusive lock.
26. Explain Two-phase Locking Protocol (2PL) and its variations.
27. What is schedule? Give example of Serial and Concurrent Schedule.
28. Write a steps of algorithm for testing serializibility of schedule.
29. What are cursors give different types of cursors.
30. Explain the purpose of checkpoint mechanism. How often should checkpoints be performed
31. What is a deadlock ? How can a deadlock occur ? explain.
32. What is two-phase locking and how does it guarantee serializability ?
33. What is database recovery? Why backups are important?
34. Explain conflict serializability with suitable example.
35. What is difference between stored procedure and stored function?
36. Write a short note on deadlock with example.

37. What is difference between OLTP and OLAP?
38. Explain Schedule based recoverability and serializibility.
39. What is database recovery? Why recovery is needed?
40. Discuss the concurrency control mechanism in detail using suitable example.
41. How Share and exclusive locks differ ?Explain.

Check whether given schedule S is conflict serializable or not
S : $R_1(A)$, $R_2(A)$, $R_1(B)$, $R_2(B)$, $R_3(B)$, $W_1(A)$, $W_2(B)$

Following is list of events in an interleaved execution of set T1, T2, and T3 assuming 2PL. Is there any deadlock? If yes, which transactions are involved in deadlock?

Time	Tansaction	Code
t1	T1	Lock (A,X)
t2	T2	Lock (B,S)
t3	T3	Lock (A,S)
t4	T1	Lock (C,X)
t5	T2	Lock (D,X)
t6	T1	Lock (D,S)
t7	T2	Lock (C,S)

Explain the deferred update technique of log based recovery following are the log entries at time of system crash

[Start_transaction, T_1]

[write_item, T_1, A, 10]

[Commit T_1]

[Start_transaction, T_3]

[Write item, T_3, B, 15]

[Checkpoint]

[Commit T_3]

[Start_Transaction, T_2]

[Write_item T_2, B, 20]

[Start_transaction, T_4]

[Write_item T_4, D, 25]

[Write_item T_2, C, 30] System crash

If a deferred update technique with check point is used, what will be recovery procedure?

Consider the following transactions. Give two non-serial schedules that are serializable.

T_1	T_2
Read (x)	Read (z)
X=x+100	Read (x)
Write (x)	X=x-z
Read (y)	Write (x)
Read (z)	Read (y)
y=y+z	y=y-100
Write (y)	Write (y)

Consider the following relation schema: **[5]**

student (sno, sname) teacher (tno, tname, qualification)

Student and teacher are related with many-many relationship.

Write a cursor to list details of students who have taken RDBMS as a subject

Consider the following relational database: **[5]**

Doctor (dno, dname, dcity) Hospital (hno, hname, hcity)

Doc-hosp (dno, hno)

Write a function to return count of number of hospitals located in 'Pune 'City.

Following is the list of events in an interleaved execution of set T_1, T_2, T_3 and T4 assuming 2PL. IS there a deadlock? If yes which transactions are involved in a deadlock?

Time	Transaction	Code
t1	T1	Lock(A,X)
t2	T2	Lock (B,X)
t3	T3	Lock (A,S)
t4	T4	Lock (B,S)
t5	T1	Lock (B,S)
t6	T3	Lock (D,X)
t7	T2	Lock (D,S)
t8	T4	Lock (C,X)

Consider the following transaction Give 2 non serial schedules that are serializable

T1	T2
Read (A)	Read (B)
A = A – 1000	B = B+100
Write (A)	Write (B)
Read (B)	Read (C)
B = B – 100	C = C +100
Write (B)	Write (C)

Consider following database

student (Sno, Sname, Sclass, Saddr)

Teacher (tno, tname, qualification, experience) The relatioship of student and teacher is M-M with descriptive attribute as subject & Marks write a trigger before deleting a student record from the student table. Raise notice and display the message “Student record is being deleted”

Consider the following schedule and draw precedence graph for that state whether schedule is serializable or not.

T1	T2
Read (A)	
A = A– 50	
	Read (A)
	temp = A* 0.1
	A = A – temp
	Write (A)
	Read (B)
Write (A)	
Read (B)	
B = B +50	
Write (B)	
	B= B + temp
	write (B)

Consider following database

Movie (mno, mname, relese-year, budget)

Actor (ano, aname, role, charges, addr)

Relationship between movie and Actor is M-M

Write a function to list moviewise charges of Amitabh Bachchan

Following are the log entries at the time of system crash:

<T1, start>

<T1, X, 40>

<T1, commit>

<checkpoint>

<T2, start>

<T2, U, 80>

<T3, start>

<T3, Z, 40>

<T2, commit>

system crash

if immediate update technique is used, what will be the recovery procedure?

The following is a list of events in an interleaved execution of set of transaction T1, T2, T3, T4 with two phase locking protocol:

Time	Transaction	Code
t_1	T1	LOCK (A,X)
t_2	T2	LOCK (B,S)
t_3	T3	LOCK (A,X)
t_4	T4	LOCK (C,S)
t_5	T1	LOCK (B,X)
t_6	T2	LOCK (C,X)
t_7	T3	LOCK (D,S)
t_8	T4	LOCK (D,S)

Find, is there any deadlock? If yes, which transactions are involved in a deadlock?

Consider the following Project-Employee database maintained by a company which stores the details of the projects assigned to the employees.

Project (pno, pname, ptype, duration)

Employee (eno, ename, qualification, joindate)

Relationship : M_M with descriptive attributes as start_date, no_of_hours_worked.

Write a stored function using cursor to accept project name as input and print the names of employees working on the project. [4]

Check whether the below schedule is conflict serializable or not, using precedence graph : [4]

S : $R_1(B)$, $R_3(C)$, $R_1(A)$, $W_2(A)$, $W_1(A)$, $W_2(B)$, $W_3(B)$, $W_1(B)$, $W_3(B)$, $W_3(C)$.

Consider the following sequence of events in an interleaved execution of transactions T_1, T_2, T_3, T_4 with 2PL protocol. Find, is there a deadlock ? If yes, which transactions are involved in a deadlock.

Time	**Transaction**	**Code**
t_1	T_1	Lock (A, X)
t_2	T_2	Lock (B, S)
t_3	T_3	Lock (A, S)
t_4	T_4	Lock (B, S)
t_5	T_1	Lock (B, X)
t_6	T_2	Lock (C, X)
t_7	T_3	Lock (D, S)
t_8	T_4	Lock (D, X)

Consider the following schedule of interleaved execution of transactions. Check if there is a deadlock in the system : [4]

T_1	T_2	T_3
		R(X)
		W(X)
	R(Y)	
R(Z)		
W(Z)		
	W(Y)	
R(X)		
W(X)		
		R(Y)
		W(Y)
	R(Z)	
	W(Z)	